STAT2004 – Tutorial Week 9 Solutions

Exercise 2 (Significance level, power, type I and type II errors)

(a) False. The significance level of a test is the probability of rejecting the null hypothesis when it is in fact true.

(b) False. If the significance level of a test is decreased, the power would be expected to decrease also.

(c) False. If a test is rejected at significance level α, then the probability that we've made a Type I error is α.

(d) False. The probability that the null hypothesis is correctly rejected is equal to the power of the test.

(e) False. A type I error occurs when the statistic falls in the rejection region of the test *and* the null hypothesis is true.

(f) Depends on your ethical standpoint. But generally false, a type I error is more serious than a type II error.

(g) False. The power of a test is determined by the distribution of the test statistic under the alternative. The level of a test is determined by the null distribution of the test statistic.

(h) True.

Exercise 3 (Testing for a biased coin)

(a) Let p be the underlying probability of heads in the given coin. Then, the null hypothesis is $H_0 : p = 0.5$ versus the alternative $H_1 : p \neq 0.5$.

(b) Let X be the number of heads from the 10 tosses of the coin. Recall that the level of the test is the probability of rejecting the null hypothesis when it is in fact true. If the null hypothesis is true, then $X \sim \text{Bin}(10, 0.5)$, and so the level of the test is

$$\begin{aligned}\alpha &= P_{H_0}(X = 0) + P_{H_0}(X = 1) + P_{H_0}(X = 9) + P_{H_0}(X = 10) \\ &= 0.5^{10} + \binom{10}{1} 0.5^{10} + \binom{10}{9} 0.5^{10} + 0.5^{10} \\ &= 0.021\,.\end{aligned}$$

(c) Recall that the power of a test is the probability of rejecting the null hypothesis when it is in fact false. Under the alternative value $p = 0.1$, the number of heads $X \sim \text{Bin}(10, 0.1)$ and so the power of the test is

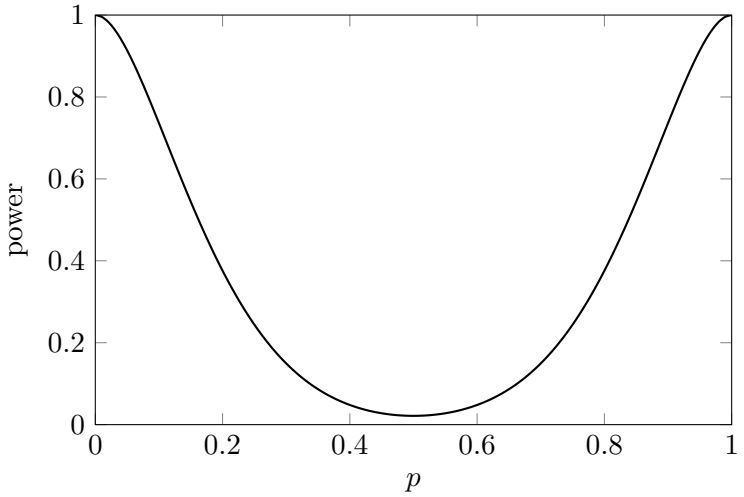
$$\begin{aligned}\text{power} &= P_{p=0.1}(X = 0) + P_{p=0.1}(X = 1) + P_{p=0.1}(X = 9) + P_{p=0.1}(X = 10) \\ &= 0.9^{10} + \binom{10}{1} 0.1^1 \times 0.9^9 + \binom{10}{9} 0.1^9 \times 0.9^1 + 0.1^{10} \\ &= 0.736\,.\end{aligned}$$

(d) We can repeat the above calculations with different alternative values of p ranging from 0 to 1. Doing this in R and plotting the power curve gives:

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#########################################################################
# create a grid of values of p from 0 to 1 in steps of 0.01
p = (0:100)/100

# power of the test "reject H0 if X = 0,1,9 or 10" evaluated at each p
power = dbinom(0,10,p) + dbinom(1,10,p) + dbinom(9,10,p) + dbinom(10,10,p)

# plot the power curve against p:
plot(p, power, type="l")
#########################################################################
```



Exercise 4* (Testing for a loaded die)

(a) Because only "sixes" are recorded, the underlying parameter here is the proportion p of "six" of this die. The null hypothesis would be $H_0 : p = 1/6$ for a fair die versus the alternative hypothesis $H_1 : p \neq 1/6$ for a loaded die.

(b) Let X be the number of sixes from 100 rolls of this die. Then under the null hypothesis $X \sim \text{Bin}(100, 1/6)$.

The one-sided p-value for testing $H_0 : p = 1/6$ versus $H_1 : p > 1/6$ is computed as $P(X \geq x_{\text{obs}}) = P(X \geq 25) = 0.022$ using software or the normal approximation to the Binomial distribution. Thus, the two-sided p-value for testing $H_0 : p = 1/6$ versus $H_1 : p \neq 1/6$ here would be $2 \times 0.022 = 0.043$.

We can therefore reject the null hypothesis in favour of the alternative at the 5% level.